Session4: Generalized Linear Models

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Outline

1. Model components
2. Generalized estimating equations (GEE) models

Systematic component

From Session 1, we learned how to model the mean of the response $\mu_i = \mathrm{E}(Y_i)$ using the linear predictor $\boldsymbol{x}_i^\top \boldsymbol{\beta}$ as $\mu_i = \boldsymbol{x}^\top \boldsymbol{\beta}$ (called linear model), where
$\boldsymbol{x} = (1, x_{i1}, \cdots, x_{ip})^\top$, $\boldsymbol{\beta} = (\beta_0, \beta_1, \cdots, \beta_p)^\top$, and $\boldsymbol{x}_i^\top \boldsymbol{\beta}$ is called the linear predictor and is the systematic component.
In the generalized linear model (GLM), we assume the systematic component is a function of the linear predictor, i.e.,

$$\begin{aligned} \mu_i &= g^{-1}(\boldsymbol{x}_i^\top \boldsymbol{\beta}) \\ \text{or alternatively} & \\ g(\mu_i) &= \boldsymbol{x}_i^\top \boldsymbol{\beta}. \end{aligned} \tag{1.1}$$

We use several references in addition to the main book, such as
Dunne, P.K. and Smyth, G.K. (2018) Generalized Linear Models With Examples in R, Springer, New York.

The components of a GLM

Any GLM has three components[1]:

The random component The response component has a distribution with mean $\mu = \mathrm{E}(\mathrm{Y})$.

The systematic component Covariates $\boldsymbol{x} = (X_1, \cdots, X_p)$ produce a linear predictor $\eta = \sum_{j=1}^{p} \boldsymbol{x}^\top \boldsymbol{\beta}$.

The link function A function which acts as a link between the random and systematic components, i.e., links μ to η.

More specifically, if $\eta = g(\mu)$, then, $g(\cdot)$ is called the link function.

[1] McCullagh, P. and Nelder, J.A. (1989) Generalized Linear Models, 2nd Ed., Chapman & Hall/CRC Monographs on Statistics and Applied Probability, New York.

Generalized linear models allow two extensions; first the distribution in component 1 may come from an exponential family other than the Normal, and secondly the link function in component 3 may become any monotonic differentiable function.

Exponential Dispersion Model (EDM)

GLMs assume the responses come from a distribution that belongs to a family of distributions called the exponential family (or EDM family, or just EDMs). Continuous EDMs include the normal, gamma, and ... distributions. Discrete case include the Poisson, binomial and negative binomial, and ... distributions. The EDM family of distributions enables GLMs to be fitted to a wide range of data types, including binary data, proportions, counts, positive continuous data, and positive continuous data with exact zeros.

EDM cont.

Distributions in the EDM family have a probability function (a probability density function if y is continuous; a probability mass function if y is discrete) of the form

$$\Pr(y;\theta,\phi) = a(y,\phi)\exp\left\{\frac{y\theta-\kappa(\theta)}{\phi}\right\}, \tag{1.2}$$

where θ is called the canonical parameter, if ϕ is known. $\kappa(\theta)$ is a known function, and is called the cumulant function. $\phi > 0$ is the dispersion parameter. $a(y,\phi)$ is the normalizing function ensuring that (1.2) is a probability function. That is, $a(y,\phi)$ is the function of ϕ and y ensuring that $\int \Pr(y;\theta,\phi)dy = 1$ over the appropriate range if y is continuous, or the function ensuring that $\sum_y \Pr(y;\theta,\phi) = 1$ if y is discrete.

EDM cont.

We have the following properties for $Y \sim EDM(\mu, \phi)$, where $\mu = \mathrm{E}(Y)$:

MGF $M_Y(t) = \exp\left\{\frac{\kappa(\theta+t\phi)-\kappa(\theta)}{\phi}\right\}$

Cumulant generating function $K_Y(t) = \log M(t) = \frac{\kappa(\theta+t\phi)-\kappa(\theta)}{\phi}$

Mean $\mathrm{E}(Y) = \mu = \frac{d\kappa(\theta)}{d\theta}$

Variance $\mathrm{Var}(Y) = \phi\frac{d^2\kappa(\theta)}{d\theta^2}$

Variance function $V(\mu) = \frac{d\mu}{d\theta} = \frac{d^2\kappa(\theta)}{d\theta^2} \implies \mathrm{Var}(Y) = \phi V(\mu)$

The variance function uniquely determines the distribution within the class of EDMs since the variance function determines $\kappa(\theta)$, up to an additive constant. This in turn specifies $K(t)$, which uniquely characterizes the distribution.

EDM cont.

EDM	$V(\mu)$	$\kappa(\theta)$	θ	ϕ
Normal	1	$\theta^2/2$	μ	σ^2
Binomial	$\mu(1-\mu)$	$\dfrac{\exp\theta}{1+\exp\theta}$	$\log\dfrac{\mu}{1-\mu}$	$\dfrac{1}{m}$
Negative binomial	$\mu+\dfrac{\mu^2}{k}$	$-\log(1-\exp\theta)$	$\log\dfrac{\mu}{\mu+k}$	1
Poisson	μ	$\exp\theta$	$\log\mu$	1
Gamma	μ^2	$-\log(-\theta)$	$-\dfrac{1}{\mu}$	ϕ
Inverse Gaussian	μ^3	$-\sqrt{-2\theta}$	$-\dfrac{1}{2\mu^2}$	ϕ
Tweedie ($\xi \leq 0$ or $\xi \geq 1$)	μ^ξ	$\dfrac{\{(1-\xi)\theta\}^{(2-\xi)/(1-\xi)}}{2-\xi}$	$\dfrac{\mu^{1-\xi}}{1-\xi}$	ϕ
		for $\xi \neq 2$	for $\xi \neq 1$	

glm function

The arguments to a glm call are as follows

glm(formula, family = gaussian, data, weights, subset, na.action, start = NULL, etastart, mustart, offset, control = glm.control(...), model = TRUE, method = "glm.fit", x = FALSE, y = TRUE, contrasts = NULL, ...)

Formula $y \sim x1 + x2$ where we can also benefit from the following symbols

- **a:b** for an interaction between a and b
- **a*b** which expands to a + b + a:b
- **-** to exclude a term or terms
- **1** to include an intercept (included by default)
- **0** to exclude an intercept

glm cont.

link functions

binomial link = "logit"

gaussian link = "identity"

Gamma link = "inverse"

inverse.gaussian link = "$1/mu^2$"

poisson link = "log"

BUT, how about Tweedie?

Tweedie GLM

In general, any EDM with a power variance function $V(\mu) = \mu^{\xi}$ is called a Tweedie distribution, or a Tweedie EDM, where ξ can take any real.

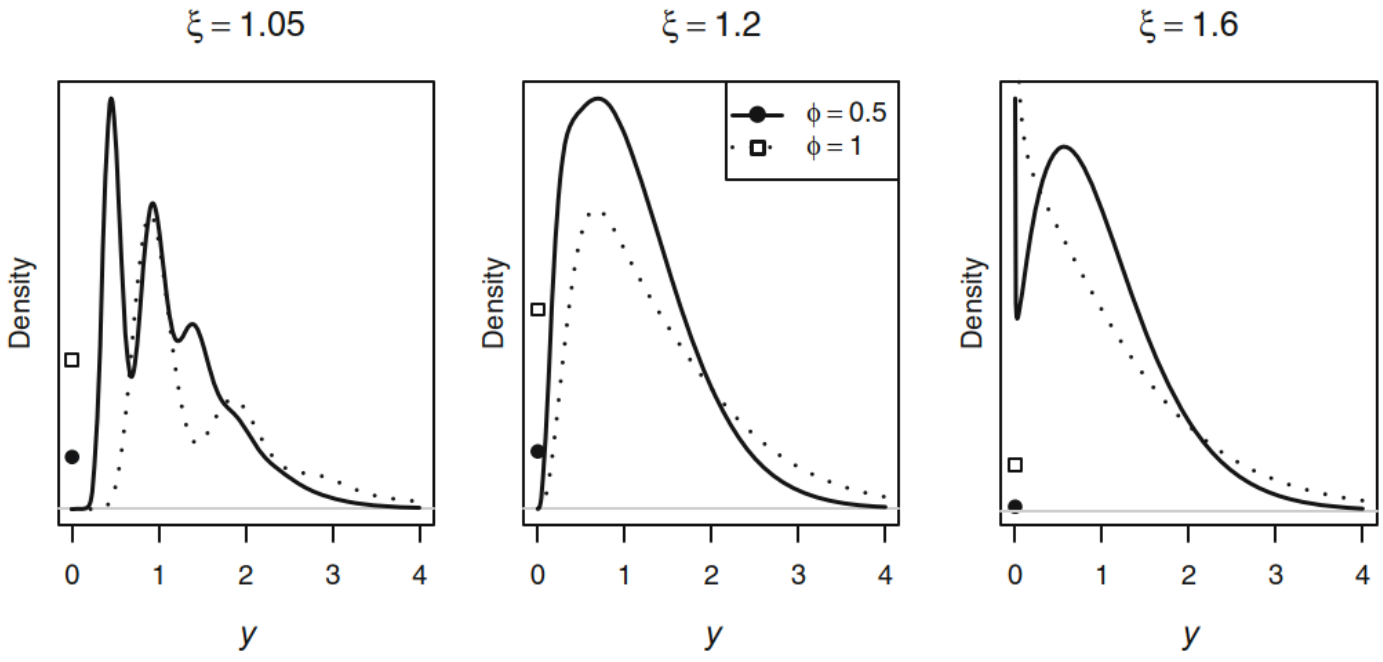
Tweedie EDM	ξ	S	Ω
Extreme stable	$\xi < 0$	$\mathbb{R}$	$\mathbb{R}^+$
Normal	$\xi = 0$	$\mathbb{R}$	$\mathbb{R}$
No EDMs exist	$0 < \xi < 1$		
Discrete	$\xi = 1$	$y = 0, \phi, 2\phi, \ldots$	$\mathbb{R}^+$
Poisson-gamma	$1 < \xi < 2$	$\mathbb{R}_0^+$	$\mathbb{R}^+$
Gamma	$\xi = 2$	$\mathbb{R}^+$	$\mathbb{R}^+$
Positive stable	$\xi > 2$	$\mathbb{R}^+$	$\mathbb{R}^+$

S is the sample space for Y (support) and Ω is the domain for μ

Tweedie GLM cont.

For $1 < \xi < 2$, the Tweedie distributions are suitable for modeling positive continuous data with exact zeros. For instance, in the rainfall modeling, when no rain falls, an exact zero is recorded, but when rain does fall, the amount is a continuous measurement.

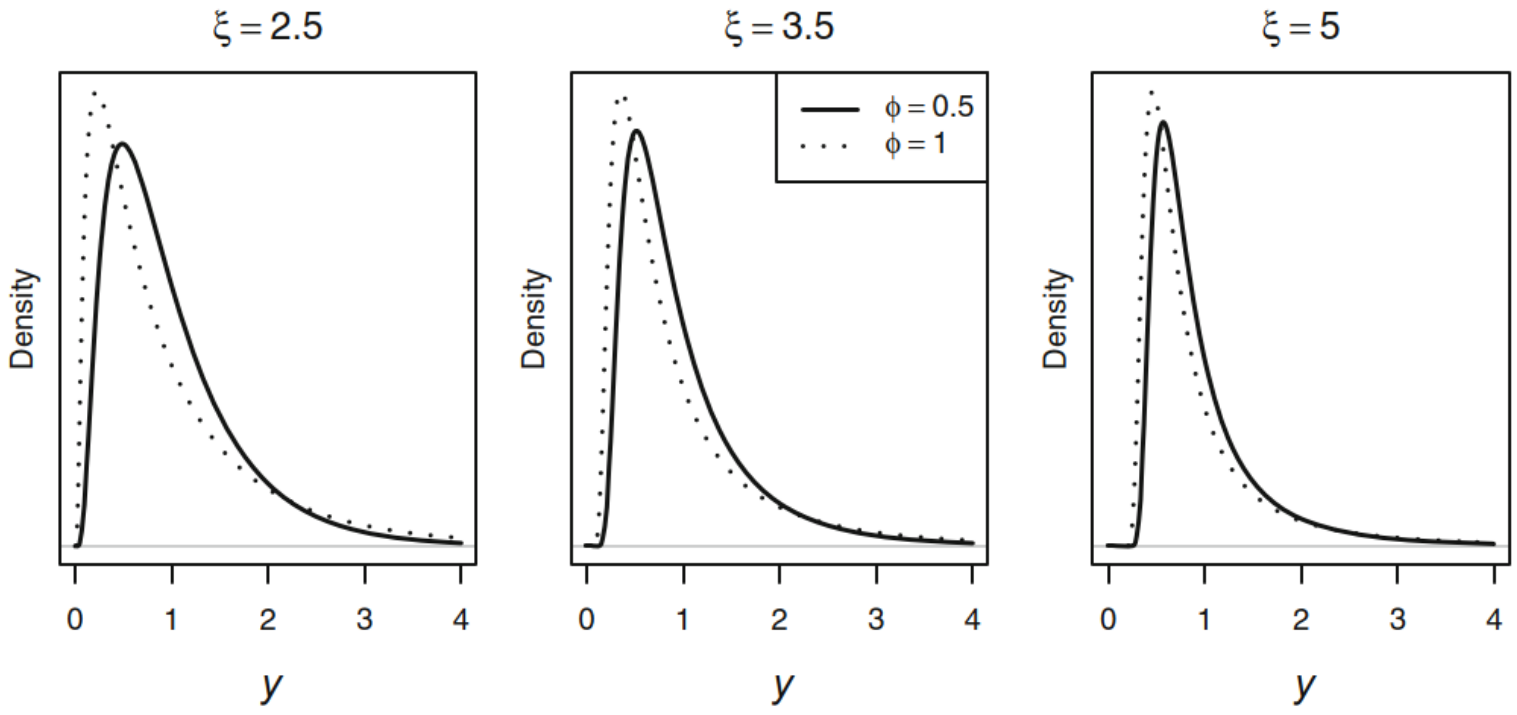
Figure: Examples of Tweedie probability functions with $1 < \xi < 2$ and $\mu = 1$. The solid lines correspond to $\phi = 0.5$ and the dotted lines to $\phi = 1$. The filled dots show the probability of exactly zero when $\phi = 0.5$ and the empty squares show the probability of exactly zero when $\phi = 1$



Tweedie GLM cont.

For $\xi \geq 2$, the Tweedie distributions are suitable for modeling positive continuous data. The gamma ($\xi = 2$) and inverse Gaussian ($\xi = 3$) distributions are special cases. The distributions become more right skewed as ξ increases.

Figure: Examples of Tweedie probability functions with $\xi > 2$ and $\mu = 1$. As ξ gets larger, the distributions become more skewed to the right. The solid lines correspond to $\phi = 0.5$; the dotted lines to $\phi = 1$



Tweedie GLM cont.

For $\xi = 1$ the Tweedie distributions are suitable for modeling discrete data where $y = 0, \phi, 2\phi, 3\phi, \cdots$. When $\phi = 2$, for example, a positive probability exists for $y = 0, 2, 4, \cdots$. The Poisson distribution is a special case when $\phi = 1$.

index parameter

ξ is called the Tweedie index parameter for the Tweedie distributions, and specifies the particular distribution in the Tweedie family of distributions.

We designate $Y \sim TW_\xi(\mu, \phi)$ for the random variable Y in the Tweedie EDM.

Lab8

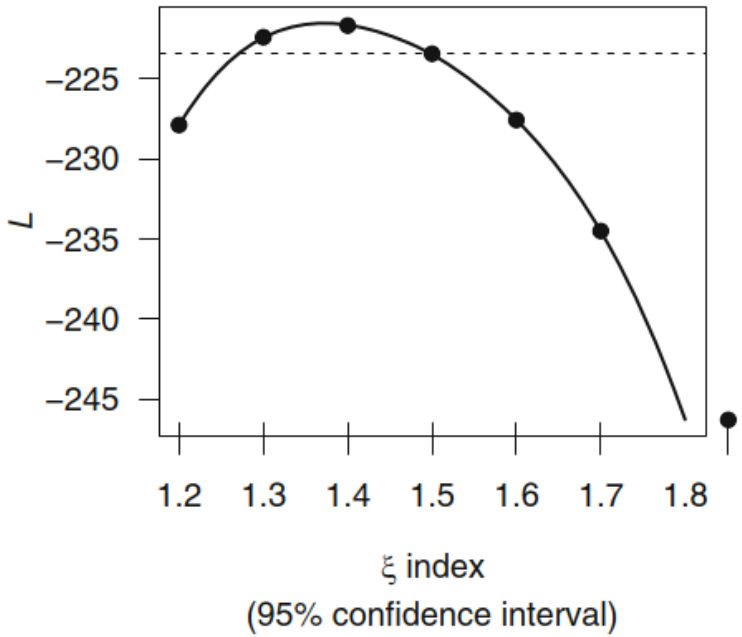
The total July rainfall (in millimeters) at Quilpie, Australia, has been recorded, together with the value of the monthly mean southern oscillation index (soi). The soi is the standardized difference between the air pressures at Darwin and Tahiti, and is known to have relationships with rainfall in parts of Australia. Some Australian farmers may delay planting crops until a certain amount of rain has fallen (a "rain threshold") within a given time frame (a "rain window"). Accordingly, we define the response variable Y as

$$Y = \begin{cases} 1 & \text{if the total July rainfall exceeds 10mm} \\ 0 & \text{o.w.} \end{cases}$$

We observe that the total monthly July rainfall is continuous, with exact zeros.

Lab8 cont.

From the output of Lab8, we have the following profile likelihood plot for the estimation of ξ.



Further, $\hat{\xi} = 1.37$ with 95% CI$=(1.27, 1.49)$.
Also $\hat{\phi} = 5.55$.

Lab8 cont.

Once an estimate of ξ has been obtained, the Tweedie GLM can be fitted in R using the usual glm() function. The Tweedie distributions are denoted in R using family=tweedie() in the glm() call, after loading the "statmod" package.

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -2.1691     1.9560 -1.1089 0.271682
Phase2        5.6923     1.9678  2.8927 0.005239 **
Phase3        3.5153     2.0600  1.7064 0.092854 .
Phase4        5.0269     1.9729  2.5480 0.013287 *
Phase5        4.6468     1.9734  2.3547 0.021665 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Lab8 cont.

The predicted number of zero-rainfall months $\hat{\pi}_0$ for each SOI phase can be compared to the actual proportion of months in the data with zero rainfall for each SOI phase.
Note that

$$\pi_0 = \Pr(Y = 0) = \exp\left\{-\frac{\mu^{2-\xi}}{\phi(2-\xi)}\right\}. \tag{1.3}$$

Using the invariance property of MLEs, the ML estimate of π_0, $\hat{\pi}_0$, say is obtained as

$$\hat{\pi}_0 = \exp\left\{-\frac{\hat{\mu}^{2-\hat{\xi}}}{\hat{\phi}(2-\hat{\xi})}\right\} \tag{1.4}$$

GLM with random effects

Recall the marginal model for the longitudinal data from Session 3. In order to constitute the random component into the systematic component of the GLM, we assume

$$\boldsymbol{\eta}_i = \boldsymbol{X}_i \boldsymbol{\beta} + \boldsymbol{Z}_i \boldsymbol{b} \tag{2.1}$$

Under the normality assumption, we specify only the first two moments of the responses for each person/individual. With normal (Gaussian) data, the first two moments fully determine the likelihood, but this is not the case with other members of the GLM family. To specify the entire likelihood, additional assumptions about higher-order moments are also necessary.

Even if additional assumptions are made, the likelihood is often intractable and involves many nuisance parameters in addition to α (here, parameters of the random effect distribution) and β that must be estimated. For this reason, a reasonable approach in many problems is to use generalized estimating equations (GEE), a multivariate analogue of quasi-likelihood (Diggle et al. (2002)[2]).

The GEE approach is an extension of the GLMs for longitudinal analysis of categorical response (discrete response); it can be also used for continuous measurements. It uses quasi-likelihood estimation and the full likelihood of the data is not specified.

It is still marginal modeling (see page 25 of Session 3: Approaches for LDA)

[2] Diggle, P., Heagerty, P., Liang, K.Y. and Zeger, S. (2002), Analysis of Longitudinal Data, Oxford University Press, UK.

GEE cont.

The additional specification in a GEE model is for the "working" correlation structure of the repeated measures. Each individual's correlation matrix $\boldsymbol{R}_i$ is of size $n_i \times n_i$. It is assumed that the correlation matrix $\boldsymbol{R}_i$, depends on a vector of association parameters denoted $\boldsymbol{\alpha}$. These parameters $\boldsymbol{\alpha}$ are assumed to be the same for all subjects (as for Sessions 2 & 3). They represent the average dependence among the repeated observations across subjects.

GEE cont. Working correlation matrix

Recall the conditional variable $\mathrm{Var}(Y_{ij}|\text{systematic component}) = \phi V(\mu_{ij})$ (see page 7).
Define $\boldsymbol{A}_i = \mathrm{diag}(V(\mu_{i1}), \cdots, V(\mu_{in_i}))$, to be a $n_i \times n_i$ diagonal matrix. Also, let $\boldsymbol{R}_i(\boldsymbol{\alpha})$ be the $n_i \times n_i$ "working correlation matrix" (of the n_i repeated measures) for the ith subject. Then, the (working) variance-covariance matrix for $\boldsymbol{Y}_i$ is given by

$$\mathrm{Var}(\boldsymbol{Y}_i) = \boldsymbol{V}_i(\boldsymbol{\alpha}) = \phi \boldsymbol{A}_i^{\frac{1}{2}} \boldsymbol{R}_i(\boldsymbol{\alpha}) \boldsymbol{A}_i^{\frac{1}{2}} \tag{2.2}$$

GEE cont.

TABLE 1: Summary of commonly used "working" correlation structures for GEE.

Correlation structure	$\mathrm{Corr}(Y_{ij}, Y_{ik})$	Sample matrix	Estimator
Independent	$\mathrm{Corr}(Y_{ij}, Y_{ik}) = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	NA
Exchangeable	$\mathrm{Corr}(Y_{ij}, Y_{ik}) = \begin{cases} 1 & j = k \\ \alpha & j \neq k \end{cases}$	$\begin{pmatrix} 1 & \alpha & \alpha \\ \alpha & 1 & \alpha \\ \alpha & \alpha & 1 \end{pmatrix}$	$\hat{\alpha} = \frac{1}{(N' - p)\phi} \sum_{i=1}^{K} \sum_{j \neq k} e_{ij} e_{ik}$ $N' = \sum_{i=1}^{K} n_i (n_i - 1)$
k-dependent	$\mathrm{Corr}(Y_{ij}, Y_{i,j+m}) = \begin{cases} 1 & m = 0 \\ \alpha_m & m = 1, 2, \ldots, k \\ 0 & m > k \end{cases}$	$\begin{pmatrix} 1 & \alpha_1 & 0 \\ \alpha_1 & 1 & \alpha_1 \\ 0 & \alpha_1 & 1 \end{pmatrix}$	$\hat{\alpha}_m =$ $\frac{1}{(K_m - p)\phi} \sum_{i=1}^{K} \sum_{j \leq n_i - m} e_{ij} e_{i,j+t}$ $K_t = \sum_{i=1}^{K} (n_i - m)$
Autoregressive AR(1)	$\mathrm{Corr}(Y_{ij}, Y_{i,j+m}) = \alpha^m,\ m = 0, 1, 2, \ldots, n_i - j$	$\begin{pmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{pmatrix}$	$\hat{\alpha} = \frac{1}{(K_1 - p)\phi} \sum_{i=1}^{K} \sum_{j \leq n_i - 1} e_{ij} e_{i,j+1}$ $K_1 = \sum_{i=1}^{K} (n_i - 1)$
Toeplitz	$\mathrm{Corr}(Y_{ij}, Y_{i,j+m}) = \begin{cases} 1 & m = 0 \\ \alpha_m & m = 1, 2, \ldots, n_i - j \end{cases}$	$\begin{pmatrix} 1 & \alpha_1 & \alpha_2 \\ \alpha_1 & 1 & \alpha_1 \\ \alpha_2 & \alpha_2 & 1 \end{pmatrix}$	$\hat{\alpha} = \frac{1}{(N' - p)\phi} \sum_{i=1}^{K} \sum_{j \neq k} e_{ij} e_{ik}$ $N' = \sum_{i=1}^{K} n_i (n_i - 1)$
Unstructured	$\mathrm{Corr}(Y_{ij}, Y_{ik}) = \begin{cases} 1 & j = k \\ \alpha_{jk} & j \neq k \end{cases}$	$\begin{pmatrix} 1 & \alpha_{12} & \alpha_{13} \\ \alpha_{21} & 1 & \alpha_{23} \\ \alpha_{31} & \alpha_{32} & 1 \end{pmatrix}$	$\hat{\alpha}_{jk} = \frac{1}{(K - p)\phi} \sum_{i=1}^{K} e_{ij} e_{ik}$

Adopted from Wang, M. (2014) Generalized estimating equations in longitudinal data analysis: A review and recent developments, Advances in Statistics, 303728. — Note e_{ij} is the Pearson residual.

GEE cont.

In the absence of a convenient likelihood function to work with, it is sensible to estimate $\boldsymbol{\beta}$ by solving a multivariate analogue of the quasi-score function (generalized estimating equation)

$$\boldsymbol{S}_{\boldsymbol{\beta}} = \boldsymbol{S}_{\boldsymbol{\beta}}(\boldsymbol{\beta}, \boldsymbol{\alpha}) = \sum_{i=1}^{n} \left(\frac{\partial \boldsymbol{\mu}_i}{\partial \boldsymbol{\beta}} \right)^{\top} \mathrm{Var}(\boldsymbol{Y}_i)^{-1} (\boldsymbol{Y}_i - \boldsymbol{\mu}_i) = \boldsymbol{0}. \tag{2.3}$$

Since usually, $\mathrm{Var}(\boldsymbol{Y}_i)$ is dependent to $\boldsymbol{\alpha}$, we can estimate both parameters by simultaneously solving $\boldsymbol{S}_{\boldsymbol{\beta}} = \boldsymbol{0}$ and

$$\boldsymbol{S}_{\boldsymbol{\alpha}} = \boldsymbol{S}_{\boldsymbol{\alpha}}(\boldsymbol{\beta}, \boldsymbol{\alpha}) = \sum_{i=1}^{n} \left(\frac{\partial \boldsymbol{\eta}_i^*}{\partial \boldsymbol{\alpha}} \right)^{\top} \boldsymbol{H}_i^{-1} (\boldsymbol{W}_i - \boldsymbol{\eta}_i^*) = \boldsymbol{0}, \tag{2.4}$$

where $\boldsymbol{\eta}_i^* = \mathrm{E}(\boldsymbol{W}_i)$ and $\boldsymbol{W}_i$ is the set of all products of pairs of residuals and squared residuals.

GEE cont.

As an example, in the normal case with marginal form, the link function is identity and thus, $\boldsymbol{\mu}_i = \boldsymbol{X}_i\boldsymbol{\beta}$ and $\boldsymbol{V}_i(\boldsymbol{\alpha})$ is the variance. In this case $\frac{\partial \boldsymbol{\mu}_i}{\partial \boldsymbol{\beta}} = \boldsymbol{X}_i$ and

$$\boldsymbol{S}_{\boldsymbol{\beta}} = \sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i(\boldsymbol{\alpha})^{-1}(\boldsymbol{Y}_i - \boldsymbol{X}_i\boldsymbol{\beta}) = \boldsymbol{0}. \tag{2.5}$$

Solving for $\boldsymbol{\beta}$, yields

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha}) = \left(\sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{V}_i(\boldsymbol{\alpha})^{-1} \boldsymbol{X}_i\right)^{-1} \sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i(\boldsymbol{\alpha})^{-1} \boldsymbol{Y}_i. \tag{2.6}$$

GEE cont.

Note that (2.6) is essentially the same as the weighted least-squares (WLS) estimator with the weight matrix $\boldsymbol{V}_i(\boldsymbol{\alpha})^{-1}$. However, in the WLS one typically knows the weight matrix, here, it depends on parameters to be estimated ($\boldsymbol{\alpha}$). In this case, the solution can proceed using iteratively reweighted least squares (IRLS) where iterative estimates of $\boldsymbol{\alpha}$ are used to yield new estimates of $\boldsymbol{\beta}$, with the procedure continuing until convergence. Because $\sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i(\hat{\boldsymbol{\alpha}})^{-1}(\boldsymbol{Y}_i - \boldsymbol{X}_i\boldsymbol{\beta}) = \boldsymbol{0}$ depends only on the mean and variance of $\boldsymbol{Y}_i$, these are quasi-likelihood estimates.

GEE cont.

Solving the GEE involves iterating between the quasi-likelihood solution for estimating $\boldsymbol{\beta}$ and a robust method for estimating $\boldsymbol{\alpha}$ as a function of $\boldsymbol{\beta}$. Essentially, it involves the following steps, which are repeated until convergence.

1. Given estimates of $\boldsymbol{V}_i(\boldsymbol{\alpha})$ and ϕ, calculate estimates for $\boldsymbol{\beta}$ using IRLS.
2. Given estimates of $\boldsymbol{\beta}$, obtain estimates of $\boldsymbol{\alpha}$ and ϕ. For this, calculate Pearson (or standardize residual) residuals

$$r_{ij} = \frac{y_{ij} - \hat{\mu}_{ij}}{\sqrt{[\boldsymbol{V}(\hat{\boldsymbol{\alpha}})]_{jj}}} \tag{2.7}$$

and use these residuals to consistently estimate $\boldsymbol{\alpha}$ and ϕ.

GEE cont.

Upon convergence, in order to perform hypothesis tests and construct confidence intervals, it is of interest to obtain standard errors associated with the estimated regression coefficients. These standard errors are obtained as the square root of the diagonal elements of the matrix $\boldsymbol{V}(\hat{\boldsymbol{\beta}})$.
The GEE provides two versions of these.

- Naive or "model-based"
- Robust or "empirical"—aka "Sandwich Estimator"

GEE cont.

Naive

$$\boldsymbol{V}(\hat{\boldsymbol{\beta}}) = \left(\sum_{i=1}^{n} \boldsymbol{D}_i^\top \boldsymbol{V}_i(\hat{\boldsymbol{\alpha}})^{-1} \boldsymbol{D}_i \right)^{-1} \tag{2.8}$$

where $\boldsymbol{D}_i = \frac{\partial \boldsymbol{\mu}_i}{\partial \boldsymbol{\beta}}$

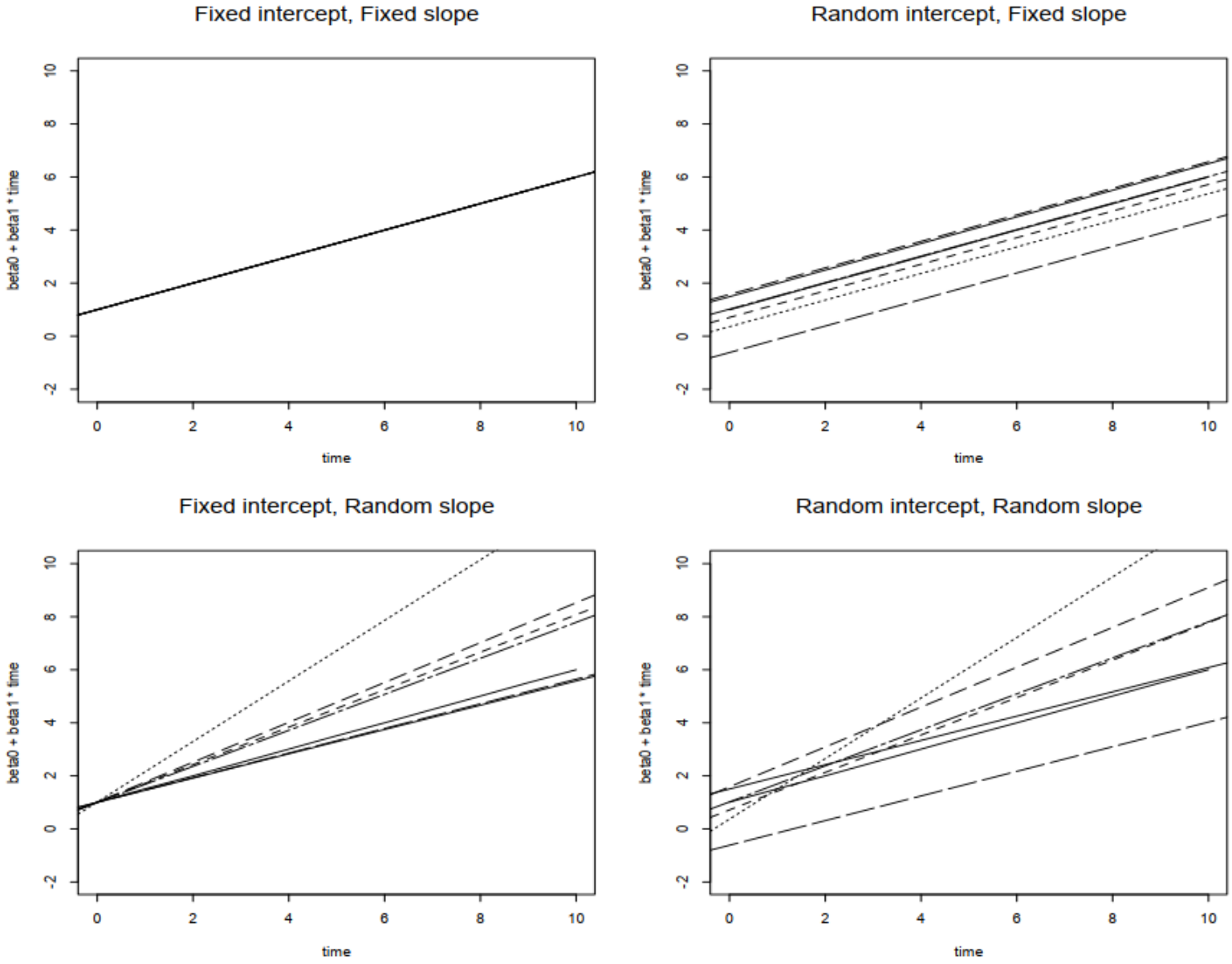
For the normal model, $\boldsymbol{V}(\hat{\boldsymbol{\beta}}) = \left(\sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i(\hat{\boldsymbol{\alpha}})^{-1} \boldsymbol{X}_i \right)^{-1}$

GEE cont.

Robust $\boldsymbol{V}(\hat{\boldsymbol{\beta}}) = \boldsymbol{M}_0^{-1}\boldsymbol{M}_1\boldsymbol{M}_0^{-1}$ where

$$\begin{aligned}
\boldsymbol{M}_0 &= \sum_{i=1}^{n} \boldsymbol{D}_i^\top \boldsymbol{V}_i(\hat{\alpha})^{-1}\boldsymbol{D}_i \\
\boldsymbol{M}_1 &= \sum_{i=1}^{n} \boldsymbol{D}_i^\top \boldsymbol{V}_i(\hat{\alpha})^{-1}(\boldsymbol{Y}_i - \hat{\boldsymbol{\mu}}_i)(\boldsymbol{Y}_i - \hat{\boldsymbol{\mu}}_i)^\top \boldsymbol{V}_i(\hat{\alpha})^{-1}\boldsymbol{D}_i
\end{aligned}$$

Before we analyze a generalized linear mixed-effects longitudinal data, let's remember



Recall two important advantages of longitudinal data: allowing us to

1. Model mean response with time and covariate (a function of mean in the context of GLM)
2. Model the variability

Three source of variability

- Heterogeneity between individuals (random effects)—because of multiple measurement on repeated individuals
- Serial correlation—two measurement closer in time will be more similar
- Measurement error

Lab9

We use the package "gee" this time.

```
gee(formula, id,
    data, subset, na.action,
    R = NULL, b = NULL,
    tol = 0.001, maxiter = 25,
    family = gaussian, corstr = "independence",
    Mv = 1, silent = TRUE, contrasts = NULL,
    scale.fix = FALSE, scale.value = 1, v4.4compat = FALSE)
```

The gee function is used in a similar way to the lme function with the addition of the features of the glm function that specify the appropriate error distribution for the response and the implied link function, and an argument to specify the structure of the working correlation matrix.

Lab9 cont. Arguments of "gee"

formula response $\sim$ predictors (systematic component)

id a vector which identifies the clusters

subset expression saying which subset of the rows of the data should be used in the fit. All observations are included by default

R a square matrix of dimension maximum cluster size containing the user specified correlation. This is only appropriate if corstr = "fixed".

tol the tolerance used in the fitting algorithm. Remember we use an iterative approach and we need to define a threshold for convergency.

family Families supported in gee are gaussian, binomial, poisson, Gamma, and quasi

corstr A character string specifying the correlation structure. The following are permitted: "independence", "fixed", "stat_M_dep", "non_stat_M_dep", "exchangeable", "AR-M" and "unstructured"

Lab9 cont.

Analyzing the "Number of Breaks in Yarn during Weaving"

A data frame with 54 observations on 3 variables.

`[,1] breaks`	numeric	The number of breaks
`[,2] wool`	factor	The type of wool (A or B)
`[,3] tension`	factor	The level of tension (L, M, H)

There are measurements on 9 looms for each of the six types of warp (`AL`, `AM`, `AH`, `BL`, `BM`, `BH`).

Lab10: R Package geepack for GEE

We start with analyzing the "Respiratory Illness" data.
The data comes from a clinical study in which the effect of a treatment of patients with respiratory illness was examined. A total of 111 patients from two clinical centers were randomized to receive either placebo or an active treatment. At four examination visits ($111 \times 4 = 444$ observations), the respiratory state of a patient was classified as good (=1) or poor (=0). Variables characterizing a patient were: center (1,2), treatment treat (A=active, P=placebo), sex (M=male, F=female), age (in years) at baseline and baseline respiratory state which is also binary. The values of the covariates were constant for the repeated elementary observations on each patient.

Lab10 cont.

A data frame with 444 observations on the following 8 variables.

center a numeric vector

id a numeric vector

treat treatment or placebo

sex M or F

age in years at baseline

baseline resporatory status at baseline

visit id of each of four visits

outcome respiratory status at each visit

Lab10 cont.

Adopted from Halekoh, U., Hojsgaard, S. and Yan, J. (2006) The R Package geepack for Generalized Estimating Equations, Journal of Statistical Software, 15(2).

	Visit	Response pattern															
	1	0	1	0	0	0	1	1	1	0	0	1	1	1	0	1	
	2	0	0	1	0	0	1	0	0	1	0	1	1	0	1	1	
	3	0	0	0	1	0	0	1	0	1	1	1	0	1	1	1	
	4	0	0	0	0	1	0	0	1	0	1	0	1	1	1	1	
Baseline	Treat																Sum
0	A	7	2	2	2	1	0	1	0	1	0	1	2	0	4	7	30
	P	18	1	0	2	1	2	0	0	1	0	0	1	2	0	3	31
1	A	0	0	0	0	0	0	1	1	0	0	4	0	1	0	17	24
	P	1	4	1	0	0	0	0	1	1	3	1	1	2	1	10	26
Sum		26	7	3	4	2	2	2	2	3	3	6	4	5	5	37	111

Table 1: Number of patients for the different response patterns classified by baseline response and treatment. The patterns are ordered according to increasing numbers of positive responses.

Do "geeglm" and "glm" give the same results? (see Assignment Q3).

Lab11: generalized linear mixed-effects model

Now, we consider mixed effect modeling for the respiratory illness dataset. First we use the "gee" package for a fixed-effects output. And then, we use the "lme4" package for the mixed-effect output. Finally, we compare the results. We will thus consider the following two models

$$\begin{aligned}
\text{M1} &= \text{logit}(\Pr(\text{outcome} = 1)) \\
&= \beta_0 + \beta_1 \text{center} + \beta_2 \text{treat} + \beta_3 \text{sex} + \beta_4 \text{age} + \beta_5 \text{baseline} \\
\text{M2} &= \text{logit}(\Pr(\text{outcome} = 1)) \\
&= \beta_0 + \beta_1 \text{center} + \beta_2 \text{treat} + \beta_3 \text{sex} + \beta_4 \text{age} + \beta_5 \text{baseline} + b
\end{aligned}$$

In M2, b is the random intercept.
Why do we need two packages for fitting the above models?

Assignments

Q1 : What is the relation between the Tweedie distribution and the Poisson-gamma distribution? Write the final model based on the estimates provided in Lab8.

Q2 : In a study of sheets of building materials the permeability of three sheets was measured on three different machines over nine days, for a total of 81 sheets, all of equal thickness. Each measurement is an average permeability of eight random pieces cut from each of the 81 sheets.
Use library(GLMsData); data(perm) to get the data.

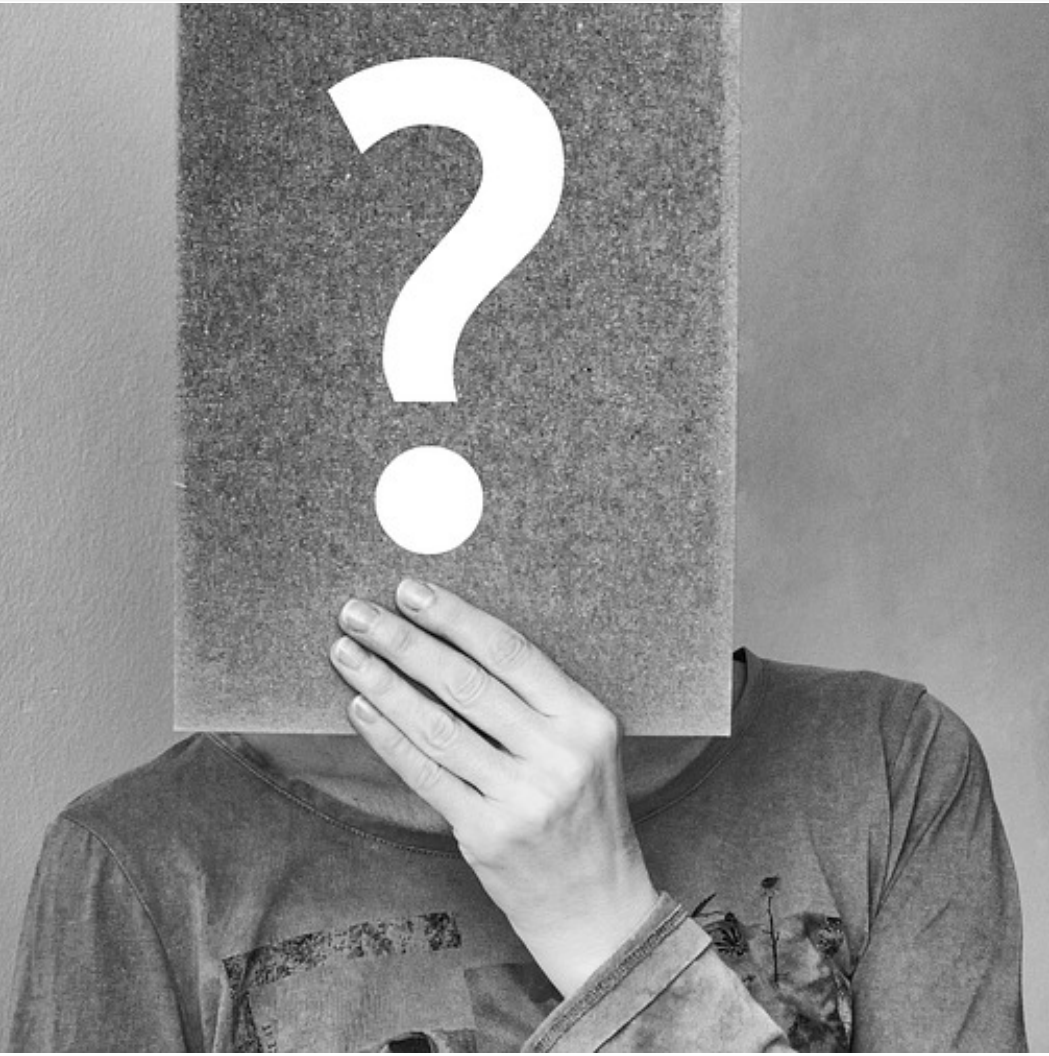
1. Determine an estimate of ξ using tweedie.profile(). What EDM is suggested?
2. Fit a suitable Tweedie GLM ensuring an appropriate diagnostic analysis.

Assignments

Q3 : Consider the respiratory illness dataset in Lab10. Run glm using

$$glm(outcome \sim center + treat + age + baseline,$$
$$data = respiratory, family = binomial())$$

Analyze the output and interpret the estimates using odds ratio, and compare the results with the output of gee.ind in the Lab10.

Q4 : In Lab11, from the output, the estimate of the scale parameter is equal to 1.015512. Thus, it is close to one and we may think of redoing the analysis by setting it to 1. Use the amendment "scale.fix = TRUE, scale.value = 1", and compare the outputs. What is the change if you apriori set the scale parameter to 1?

Thank you for your attention.